

EXPERIMENT 3 – System Calls: Fork, Exit, Getpid, Wait, Close

Aim: To study and implement process management system calls: `fork()`, `getpid()`, `wait()`, `exit()`, and `close()`.

Program 1 – Process Creation using fork(), getpid(), wait(), exit()

C Program:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    pid = fork();

    if(pid < 0)
    {
        printf("Fork Failed\n");
        exit(1);
    }
    else if(pid == 0)
    {
        printf("\nCHILD PROCESS");
        printf("\nChild PID : %d", getpid());
        printf("\nParent PID : %d\n", getppid());
        exit(0);
    }
    else
    {
        wait(NULL);
        printf("\nPARENT PROCESS");
        printf("\nParent PID : %d", getpid());
        printf("\nParent's Parent PID : %d\n", getppid());
    }

    return 0;
}
```

Compilation: `gcc fork\_demo.c -o fork\_demo`

Execution: `./fork\_demo`

Output:

```
CHILD PROCESS
Child PID : 982
Parent PID : 981
```

```
PARENT PROCESS
Parent PID : 981
Parent's Parent PID : 979
```

Shell Script Equivalent

```
#!/bin/bash

echo "Parent Process ID : $$"

(
    echo "Child Process ID : $$"
    echo "Parent Process ID : $PPID"
    exit 0
) &

wait

echo "Child Process Completed"
```

Output:

```
Parent Process ID : 983
Child Process ID : 983
Parent Process ID : 979
Child Process Completed
```

Program 3 – close() System Call

C Program:

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int fd;

    fd = open("sample.txt", O_RDONLY);

    if(fd < 0)
    {
        printf("File Opening Failed\n");
        return 1;
    }

    printf("File Opened Successfully\n");

    close(fd);

    printf("File Closed Successfully\n");

    return 0;
}
```

Output:

```
File Opened Successfully
File Closed Successfully
```

Shell Script:

```
#!/bin/bash

exec 3< sample.txt

echo "File Opened Successfully"

exec 3<&-

echo "File Closed Successfully"
```

Result: Process management system calls were studied and implemented successfully.
